

Koundinya Pidaparthi

Full Stack Software Engineer

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Full Stack Software Engineer with 3+ years of experience building scalable web applications. Proficient in React, Next.js, Node.js, TypeScript, and cloud platforms (AWS, GCP). Passionate about crafting performant, user-first products and integrating AI-powered features.

EXPERIENCE

Anchor Operating System

June 2025 – Present

Software Engineer

New York, NY (Remote)

- Architected and shipped core modules of an AI-first operating system layer using Next.js 14 App Router, TypeScript, and React Server Components, reducing time-to-interactive by 38%.
- Designed and implemented a real-time collaboration engine using WebSockets and Redis pub/sub, enabling sub-100ms latency for multi-user document editing across distributed sessions.
- Built a plugin/extension marketplace infrastructure supporting dynamic module loading via Module Federation, cutting third-party integration time from weeks to days.
- Established CI/CD pipelines with GitHub Actions and AWS CDK, automating deployments to ECS Fargate with zero-downtime blue/green releases.
- Led the migration of a monolithic REST API to a GraphQL federation layer (Apollo Router + subgraphs), improving query performance by 45% and eliminating N+1 issues.
- Mentored two junior engineers through weekly 1:1s, code review culture, and pair programming sessions, accelerating their onboarding by 30%.

Tech: Next.js 14 · TypeScript · React Server Components · WebSockets · Redis · GraphQL · Apollo · AWS CDK · ECS Fargate · GitHub Actions

Anchor Operating System

May 2024 – Sep 2024

Software Engineer Intern

New York, NY (Remote)

- Contributed to the early-stage product by building the core dashboard UI using React, Zustand, and Tailwind CSS, establishing the design-system foundation adopted by the full team.
- Implemented OAuth 2.0 authentication flow with NextAuth.js supporting Google and GitHub providers, reducing login friction and improving conversion by 22%.
- Developed RESTful API endpoints in Node.js/Express integrated with MongoDB Atlas, powering the user preferences and settings modules.
- Wrote unit and integration tests with Jest and React Testing Library, achieving 82% code coverage on critical user-flow components.
- Collaborated with the design team to translate Figma prototypes into pixel-perfect, accessible (WCAG 2.1 AA) React components using Framer Motion for micro-interactions.

Tech: React · Zustand · Tailwind CSS · NextAuth.js · Node.js · Express · MongoDB Atlas · Jest · Framer Motion

Hornblower Group

Aug 2022 – Jul 2023

Software Developer

New York, NY

- Built and maintained customer-facing ticketing and event management features using React, Redux, and GraphQL, serving over 2 million annual users across ferry and dining experience platforms.
- Optimized SQL and MongoDB query patterns, reducing average API response time from 850ms to 210ms for the high-traffic booking service.
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Integrated Stripe payment gateway with idempotent transaction handling, processing \$5M+ in monthly bookings with a 99.98% success rate.

- Developed a seat-selection microservice in Node.js/TypeScript backed by DynamoDB with DynamoDB Streams for real-time availability updates.
- Implemented feature flags with LaunchDarkly enabling gradual rollouts to 5% ' 100% of users, reducing production incidents by 60%.
- Participated in on-call rotation, triaging and resolving P1/P2 incidents for the ticketing platform with an average MTTR of under 25 minutes.

Tech: React · Redux · GraphQL · Node.js · TypeScript · MongoDB · DynamoDB · Stripe · LaunchDarkly · AWS Lambda

Syscloud Technologies

June 2021 – June 2022

Frontend Software Engineer Intern

Hyderabad, India

- Developed responsive SaaS dashboard components in React and Material UI for Google Workspace backup and recovery products, improving user task completion rate by 18%.
- Built reusable component library with Storybook documentation, standardizing UI patterns across 4 product teams and reducing duplicate code by 35%.
- Integrated Google Drive and Gmail APIs to display backup status, quota usage, and restore progress with real-time polling using React Query.
- Implemented lazy loading, code splitting, and image optimization strategies, cutting initial bundle size from 1.8MB to 620KB and improving Lighthouse score from 54 to 89.
- Collaborated in an Agile/Scrum environment with two-week sprints, participating in daily standups, sprint planning, and retrospectives.

Tech: React · Material UI · Storybook · React Query · Google Workspace APIs · JavaScript · CSS3 · Webpack

PROJECTS

AplifyAI

Jan 2025 – May 2025

Next.js 14 · TypeScript · OpenAI GPT-4 · LangChain · Pinecone · Supabase · Tailwind CSS · Stripe

An AI-powered job application platform that auto-tailors resumes and cover letters to specific job descriptions using RAG and GPT-4.

- Built a Retrieval-Augmented Generation (RAG) pipeline using LangChain and Pinecone vector database to semantically match resume sections to job descriptions, improving relevance scores by 60%.
- Integrated OpenAI GPT-4 API to generate context-aware, ATS-optimized resume bullets and cover letters in under 8 seconds per request.
- Designed a multi-tenant SaaS architecture on Supabase (PostgreSQL + Row Level Security) with Stripe billing supporting Free, Pro, and Enterprise tiers.
- Implemented a real-time PDF editor using @react-pdf/renderer with live preview, enabling users to see AI edits side-by-side before exporting.
- Achieved 98 Lighthouse performance score through Next.js App Router streaming, React Suspense boundaries, and edge caching on Vercel.

Harmony AI

Aug 2024 – Dec 2024

React · Node.js · Express · OpenAI Whisper · GPT-4 · MongoDB · AWS S3 · Socket.io

An AI-driven music collaboration platform that transcribes, analyzes, and provides real-time feedback on musical performances using speech-to-text and LLMs.

- Built a real-time audio transcription pipeline using OpenAI Whisper API, processing audio streams in 200ms chunks to deliver sub-second transcription latency.
- Developed an LLM-based music feedback engine with GPT-4 that analyzes tempo, melody, and lyrics from transcribed sessions, providing structured coaching suggestions.
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Implemented WebSocket rooms via Socket.io enabling live collaboration between multiple musicians, with cursor presence, synchronized playback, and conflict-free edits.

- Engineered audio file upload and storage pipeline using AWS S3 pre-signed URLs, reducing server load by offloading uploads directly from the browser.
- Designed a session history and analytics dashboard with MongoDB aggregation pipelines, surfacing practice trends and progress over time.

AI Exam Prep Platform

Mar 2024 – Jul 2024

Next.js · TypeScript · OpenAI GPT-4 · Prisma · PostgreSQL · Tailwind CSS · React Query · Vercel

An adaptive exam preparation platform that generates personalized practice questions, evaluates answers, and tracks mastery using AI.

- Developed an adaptive question-generation engine powered by GPT-4 that dynamically adjusts difficulty based on a student's historical performance using a spaced-repetition algorithm (SM-2).
- Built a natural-language answer evaluation system that compares free-text student responses to model answers and returns a structured rubric score with GPT-4.
- Designed the database schema with Prisma and PostgreSQL supporting multi-subject topic trees, question banks, and per-user attempt history with efficient indexing.
- Implemented a study-mode dashboard with real-time performance analytics using React Query and Chart.js, visualizing accuracy trends per topic and time spent.
- Deployed on Vercel with edge functions for AI inference endpoints, achieving p95 latency of under 1.2 seconds for question generation.

SKILLS

Languages: JavaScript (ES2022+), TypeScript, Python, Java, HTML5, CSS3, SQL, GraphQL

Frontend: React, Next.js, React Native, Redux, Zustand, Framer Motion, Tailwind CSS, Material UI, shadcn/ui, Storybook, Webpack, Vite

Backend: Node.js, Express.js, GraphQL (Apollo Server), REST APIs, WebSockets, NextAuth.js, Prisma, Redis

Databases: MongoDB, PostgreSQL, DynamoDB, MySQL, Firebase Firestore, BigTable, Redis

Cloud & DevOps: AWS (EC2, Lambda, S3, DynamoDB, CDK, API Gateway, ECS, IAM), GCP (GCS, BigTable, Cloud Build, Cloud Run), GitHub Actions, Docker, Terraform, Vercel, Netlify, CI/CD Pipelines

Testing: Jest, React Testing Library, Cypress, Vitest, Playwright

Tools & Practices: Git / GitHub, Jira, Figma, Postman, LaunchDarkly, Stripe, Agile / Scrum, Code Review, System Design

EDUCATION

Pace University

May 2025

Master of Science in Computer Science · GPA: 3.95

New York, NY

- Dean's List — every semester
- Graduate Teaching Assistant — Data Structures & Algorithms

Jawaharlal Nehru Technological University

May 2021

Bachelor of Technology in Computer Science & Engineering

Hyderabad, India